ISE-A1

Problem and Stakeholder Evidence Brief

Two-Stage Clinic Appointment and Arrival-Confirmation Workflow for Non-Emergency Outpatient Care in China

Course: Intelligent Software Engineering
Assessment: ISE-A1
Maximum: 5 marks
Evidence reviewed for this brief: 2 October 2026

Project focus: Investigate whether a two-stage workflow - online appointment allocation to a time window, followed by on-site arrival confirmation and queue registration - can reduce avoidable waiting and improve use of appointment capacity without making access harder for patients.

This brief uses public evidence from Chinese outpatient settings. It does not claim private interviews, unpublished clinic observations, or patient-level data that were not actually collected.

1. Result Summary

This brief investigates a workflow problem in non-emergency outpatient care in China: the mismatch between an appointment that is booked and the patient who actually arrives at the clinic. Published Chinese studies provide evidence for three connected pressures. Long waiting and registration queues have been documented in outpatient care; reservation services have been associated with shorter waiting; and missed online appointments remain a measurable source of unused capacity. For example, a retrospective study of 48,777 online appointments in a general hospital in Jinan reported a 15% no-show rate, while a study at Guangzhou Women and Children's Medical Center reported a large increase in appointment registration and a significant reduction in waiting time after a comprehensive reservation service was introduced [1,2]. National guidance also promotes timed appointment booking, real-name registration, no-show management and closer integration of online and in-person services [3,4]. The proposed intervention is a two-stage workflow: the patient books online into a time window such as 09:00-11:00, then confirms physical arrival at a kiosk or staffed check-in point. Check-in changes the booking state to present and places the patient in a managed queue. The clinic can then see confirmed arrivals without repeatedly checking manually. The evidence does not prove that this design will outperform existing systems in every clinic. A Shanghai survey found low use of one web appointment system and continued demand for on-the-spot booking, highlighting adoption and accessibility risks [5]. The semester project therefore focuses on a bounded prototype and measurable evaluation rather than a claim of universal improvement.

Proposed research question: How could a two-stage appointment and arrival-confirmation workflow affect patient waiting time, appointment-capacity utilisation and manual queue-coordination workload in a selected non-emergency outpatient clinic setting in China?

2. Problem, Context and Authentic Workflow

The problem is located in non-emergency outpatient care. The setting is deliberately narrower than “all clinics in China” because clinics differ in size, specialty, staffing, digital maturity and patient population. The semester intervention is therefore framed as a clinic-level scheduling and check-in prototype for a non-emergency outpatient workflow that already has some appointment capacity and a reception/check-in point.

Chinese public health policy has progressively promoted appointment-based and online/offline outpatient services. The National Health Commission reported in March 2024 that 82.7% of secondary-or-above public hospitals were providing appointment services and that tertiary public hospitals had an average appointment rate of 61.1% [6]. The same policy direction emphasises timed appointments, shorter waiting, real-name booking, and management of cancellations and missed appointments [3,4]. These facts make appointment coordination a relevant software-engineering problem, while also showing that the intervention is not intended to replace a process that does not exist.

Earlier empirical work helps reconstruct the workflow. At West China Hospital, a 2012 survey described six appointment channels, including on-site, telephone, doctor-mediated, bank-terminal

and web-based registration. Patients who could not use the systems or did not have the required health card were less able to access the digital channels [7]. In Shanghai, a 2014 survey of 1,000 outpatients reported that patients were least satisfied with waiting time; only 17% used the web appointment system studied, and many surveyed doctors still supported an on-the-spot option [5].

2.1 Current-state workflow model

Step	Current workflow (evidence-informed model)	Main uncertainty / pain point
1. Booking	Patient books in person, by telephone, web/mobile channel, or through a clinic-assisted route, depending on the setting [5,7].	Different channels can fragment information and create unequal access.
2. Appointment allocation	A clinic or doctor releases appointment capacity and assigns a date/time or appointment window [3,4].	The allocated time may not reflect later arrival or demand variation.
3. Patient travel	Patient travels to the clinic for the booked consultation.	Traffic, public transport, family duties and other delays can affect arrival time.
4. Arrival / registration	Patient may need to register, take a number or confirm arrival through a front-desk or self-service process.	The booking record and physical presence can become temporarily out of sync.
5. Queue management	Reception/clinic staff coordinate who is waiting and who is ready for the doctor.	Manual checking can consume staff attention and can be difficult during peaks.
6. Consultation	Doctor sees the patient according to the clinic's queue and clinical rules.	Exact queue order may depend on appointment type, doctor availability or urgency; software must not assume simple first-come-first-served care.

Problem definition: The specific problem is not simply “clinics need online booking”. It is the operational gap between (a) a reserved appointment, (b) the patient's actual arrival, and (c) the clinic's ability to manage confirmed arrivals efficiently. The proposed system tries to make that state transition explicit.

3. Evidence Ledger

The evidence base intentionally combines independent source types: official Chinese government guidance/statistics and peer-reviewed empirical studies. This reduces dependence on a single source type and allows the project claim to be challenged from both operational and policy perspectives.

ID	Source type	Source	Evidence extracted	Implication for project	Limit / challenge
E1	Official policy / NHC	National Health Commission, 2020 notification on appointment systems and smart-hospital construction [4].	Secondary+ hospitals should provide timed outpatient appointments; the guidance calls for real-name booking, source management, and shorter in-hospital waiting.	Supports the relevance of timed appointment and online/offline workflow integration.	Policy is national guidance; it does not measure this project's target clinic or prove a particular design will work.
E2	Official policy / NHC	NHC 2023 “Improve medical experience” campaign [3].	Promotes multiple appointment routes, time-sliced booking, real-name registration, no-show management, and redesign of outpatient processes to reduce waiting.	Directly aligns with the project's focus on arrival confirmation and no-show handling.	The policy gives direction, not a comparative experiment between specific software designs.
E3	Official national statistics / NHC	NHC press briefing, 19 March 2024 [6].	Reported 82.7% of secondary+ public hospitals with appointment services and 61.1% average appointment rate for tertiary public hospitals.	Shows that appointment systems are already a substantial part of outpatient operations, so the project should integrate rather than replace existing practices.	Published in 2024; not a 2026 census and not representative of all clinic types.
E4	Peer-reviewed empirical study	Yu et al., West China Hospital survey, 1,009 outpatients, 2012 [7].	63.4% used appointment systems; on-site and telephone channels were still substantial; lack of capability and lack of a registered health card were barriers.	Supports multi-channel access and the need for a non-digital fallback.	Older data and a major hospital; behaviour may differ in 2026 or in smaller clinics.
E5	Peer-reviewed empirical study	Zhang et al., Shanghai public hospital survey, 1,000 outpatients + 197 doctors, 2014 [5].	Patients were least satisfied with waiting time; only 17% used the studied web appointment system; doctors still saw value in alternative/on-the-spot booking.	Challenges the assumption that a software-first workflow will automatically be adopted.	One hospital and an older study; usage depends on local system design and context.
E6	Peer-reviewed empirical study	Xie et al., Guangzhou Women and Children's Medical Center, 2019 [1].	After a comprehensive reservation service, appointment registration increased from about 35% to about 89%; waiting time and perceived need to improve waiting time decreased significantly.	Shows that improving reservation infrastructure can be associated with better operational and patient outcomes.	Single Grade 3A hospital and the study design does not isolate the effect of every individual feature.

ID	Source type	Source	Evidence extracted	Implication for project	Limit / challenge
E7	Peer-reviewed empirical study	Retrospective online-appointment analysis, Jinan general hospital, 2017-2018 [2].	48,777 online appointment records contained a 15% no-show rate; no-shows varied with several patient/appointment factors.	Provides a concrete published baseline showing that online booking does not remove the no-show problem.	The 15% figure is a published reference baseline, not a measurement of the target clinic.
E8	Peer-reviewed modelling study	Luo et al., outpatient scheduling-window optimisation using West China Hospital data [8].	The choice of scheduling window affects waiting/idle-time trade-offs; the magnitude of the window matters.	Directly cautions against choosing two- or three-hour windows without examining clinic demand and capacity.	A modelling study focused on a particular hospital context; a prototype still needs local validation.

4. Stakeholder and Workflow Map

The stakeholder model separates people who directly use the system from people who are affected by its outcomes. Roles and responsibilities below are an evidence-informed synthesis of outpatient appointment workflows; a real pilot would validate the exact responsibilities with the target clinic.

Stakeholder	Role in current workflow	Need / pain point	Responsibility in proposed workflow	Influence
Patient	Requests an appointment, travels to clinic, checks in and waits.	Convenience, predictable arrival, fair queue treatment, understandable instructions.	Book a window; arrive within window; confirm arrival; receive queue status.	High personal impact; limited control over clinic capacity.
Reception / registration staff	Manage appointments, registration, arrivals and patient flow.	Manual checking and repeated status questions can compete with other front-desk work.	Monitor booking/check-in states; assist exceptions; handle fallback/manual check-in.	High operational influence.
Doctor / clinical team	Provides consultation and may have appointment-specific constraints.	Uncertain arrival status can make the schedule less predictable.	See the confirmed queue and mark consultation state; retain clinical control over order.	High service impact.
Clinic manager	Sets capacity, appointment rules and operational policies.	Needs good utilisation without creating queue overload or unfair access.	Configure windows, capacity, no-show/late rules and monitor performance.	High decision authority.
IT / system administrator	Maintains service, devices and access rights.	Needs reliable deployment, logs, recovery and low support burden.	Maintain users, devices, backups, monitoring and access controls.	High technical influence.
Patient support / accessibility staff	Help users who cannot use digital or kiosk channels easily.	Digital exclusion can create barriers for older or less capable users.	Provide assisted booking and an alternative check-in route.	Important for equitable access.

4.1 Proposed two-stage workflow

Stage	Patient action	System / staff action	State transition
1. Online booking	Select clinic/service and available date/window; confirm identity as required.	Create a booking with a unique reference and the selected window.	Available -> Booked
2. Pre-arrival reminder	Review the window and instructions; cancel or reschedule where allowed.	Send reminder; update status if patient cancels.	Booked -> Cancelled / still Booked
3. Physical arrival	Enter clinic and use kiosk/desk to confirm presence.	Verify booking, prevent duplicate check-in, issue queue number.	Booked -> Checked-in
4. Waiting	Wait for call/display/notification.	Show confirmed queue and exceptions to authorised staff.	Checked-in -> Waiting
5. Consultation	Enter consultation when called.	Mark patient as in-service and then completed.	Waiting -> In-service -> Completed
6. Exception	Arrive late, fail to arrive, lose booking reference or need assisted service.	Apply policy; preserve audit trail; use fallback route.	Booked / Checked-in -> Exception state

5. Pain, Baseline, Value Measure and Failure Cost

Because no private target-clinic dataset or field observation was supplied for this assessment brief, the project uses published Chinese evidence as a reference baseline and defines the exact measures that a later pilot or simulation should collect. This avoids presenting literature figures as if they were local clinic measurements.

Measure	Published/reference baseline	How the project would measure it	Why it matters
No-show rate	15% in a 48,777-record online-appointment dataset from a Jinan general hospital [2].	No-show appointments / all booked appointments for the evaluation period.	Shows booked capacity can remain unused.
Appointment-system use	17% of patients used the studied web appointment system in a 2014 Shanghai hospital survey; 63.4% used appointment systems in the West China Hospital survey, illustrating context dependence [5,7].	Share of bookings by channel and share of eligible patients using the online path.	Adoption is a design variable, not an automatic outcome.
Waiting time	Reservation interventions in Guangzhou were associated with a significant reduction in waiting time [1].	Arrival/check-in timestamp to consultation-start timestamp; report median and upper percentile.	Core patient-experience outcome.

Measure	Published/reference baseline	How the project would measure it	Why it matters
Window compliance	No project-specific published value identified.	Share arriving within selected window; distribution of early/late arrivals.	Tests whether flexible windows are operationally usable.
Manual coordination workload	No project-specific published value identified.	Count staff interventions, manual look-ups, calls, queue corrections per 100 appointments.	Tests the claimed admin benefit.
Capacity utilisation	No project-specific published value identified.	Completed consultations / released capacity; separately report no-show capacity.	Tests whether appointment slots are used effectively.

A useful illustration is the published 15% no-show reference rate. If a hypothetical clinic released 100 booked slots and experienced the same rate, roughly 15 bookings would be expected to be missed. This is an illustration derived from the published percentage, not a prediction for the target clinic. The project should measure the local rate before using it for capacity planning.

Failure cost has both operational and patient dimensions. A missed appointment can leave clinician capacity unused; other patients may face longer access delays; and staff may spend time calling, checking or manually reallocating patients. Conversely, a badly designed flexible-window system could make queues worse by concentrating arrivals. The value claim is therefore conditional: the intervention is worthwhile only if it reduces avoidable coordination work and/or improves useful utilisation without creating unacceptable waiting or access barriers.

6. Software Intervention versus Non-Software Alternative

The proposed software should not be accepted merely because the problem involves information. At least one non-software response must be considered. The baseline alternative below is an operational-policy change using existing channels and staff practice.

Criterion	Two-stage software workflow	Non-software operational change
Core idea	Online booking into a flexible window + on-site arrival confirmation + digital queue state.	Use fixed or flexible windows, standardised receptionist check-in, reminder calls/messages and a manual waitlist.
Patient experience	Potentially fewer rigid “exact minute” expectations and clearer arrival status.	Can also provide flexible windows but may rely more on staff communication.
No-show handling	System can identify unconfirmed arrivals and apply a configurable waitlist/release policy.	Receptionists identify no-shows and release capacity manually.
Staff workload	Potentially lower repeated checking once the workflow is adopted.	Lower technology cost but continued manual coordination.
Implementation cost	Higher: software, kiosk/device, testing, training and maintenance.	Lower: mainly policy design, scripts, signage and staff training.
Accessibility risk	Digital/kiosk use can exclude some patients unless assisted/fallback routes exist [5,7].	Can preserve human assistance more easily, but may be slower and less transparent.

Criterion	Two-stage software workflow	Non-software operational change
Data and failure risk	Requires reliable identity/booking data, access control, uptime and recovery.	Fewer technical failure modes but more dependence on human consistency.
Evidence needed	Pilot or simulation comparing waiting, no-show capacity, workload and window compliance.	Baseline measurement to determine whether operational change alone is sufficient.

Engineering decision at A1: Proceed to the semester project only as a bounded prototype investigation. The A1 evidence supports the problem class and makes the proposed workflow testable, but it does not yet establish that the software is superior to a simpler operational change in every clinic.

7. Semester Scope, Non-Goals, Assumptions and Stop Conditions

7.1 Proposed semester scope

- One non-emergency outpatient clinic context, with one to three service types rather than a hospital-wide rollout.
- Patient-facing booking flow that allocates a patient to a configurable time window, with a prototype default of two or three windows per day depending on capacity.
- Arrival confirmation through a kiosk or equivalent check-in screen, with an assisted/manual fallback for patients who cannot use the kiosk.
- Clinic staff dashboard showing booked, checked-in, waiting, in-service and completed states.
- Basic queue-number generation and notification/display support; queue ordering remains governed by clinic policy and is not treated as an automated clinical decision.
- No-show, cancellation and late-arrival states with configurable policies and an auditable status history.
- Evaluation using simulation and/or a controlled prototype trial with synthetic test data unless a real clinic provides permission and data.

7.2 Non-goals

- No diagnosis, triage, prescription recommendation or automated clinical decision making.
- No full electronic medical record, billing, insurance or inter-hospital integration.
- No emergency-department scheduling.
- No guarantee of shorter clinical consultation time; the project measures appointment/queue coordination rather than clinical treatment performance.
- No requirement to replace existing booking channels; the prototype should coexist with assisted or alternative routes.

7.3 Assumptions

- The target workflow is for planned, non-emergency outpatient visits.
- The clinic has a defined daily capacity or can provide enough information to model it.
- A physical arrival-confirmation point is available or can be simulated.
- The clinic can define what counts as on-time, late, cancelled and no-show.

- Patients can access at least one booking channel, but the design must include a human-assisted fallback.

7.4 Proposed stop / reconsider conditions

- The selected clinic workflow has no meaningful separation between booking and arrival status, so the two-stage intervention adds little operational value.
- The available evidence shows that the main bottleneck lies elsewhere (for example doctor capacity), and the prototype cannot reasonably influence it within the semester.
- Simulation repeatedly shows that the chosen window design creates unacceptable arrival bunching or longer queues than the baseline workflow.
- The clinic cannot provide an acceptable non-digital or assisted path, creating a material access barrier for patients who cannot use the digital route.
- Required operational data cannot be obtained or safely synthesised, preventing reproducible evaluation.

8. Normal, Boundary and Abuse Scenarios

8.1 Normal scenarios

ID	Scenario	Expected behaviour / acceptance
N1	Patient books 09:00-11:00 window and arrives at 09:35.	Booking is found; patient checks in once; queue number is issued; staff dashboard changes to Checked-in/Waiting.
N2	Patient cancels before the window starts.	Booking becomes Cancelled; released capacity becomes visible according to clinic policy; audit record retained.
N3	Doctor calls the next eligible patient.	Queue state changes to In-service; the patient cannot be simultaneously shown as waiting.
N4	Patient completes consultation.	State changes to Completed and the appointment is no longer considered available queue capacity.

8.2 Boundary scenarios

ID	Scenario	Expected behaviour / acceptance
B1	Patient arrives five minutes before the start of the window.	Check-in policy handles early arrival consistently; the patient is not treated as late, but queue entry may depend on clinic policy.
B2	Patient arrives five minutes after the window closes.	System marks Late / Exception and applies the clinic's published late-arrival policy; no silent deletion of the appointment.
B3	The morning window is heavily oversubscribed because many patients arrive near 09:00.	Dashboard highlights load; staff can see the queue; the system must not claim the window itself prevents congestion.

ID	Scenario	Expected behaviour / acceptance
B4	Patient cannot use the kiosk.	Assisted check-in can be completed by authorised staff without forcing the patient to abandon the appointment.
B5	Kiosk or network is unavailable.	Fallback manual check-in is used; events are reconciled after recovery; no data are silently lost.
B6	Patient has no booking reference but can provide the minimum permitted identity details.	Staff follow a verified recovery path; the system does not expose another patient's booking.

8.3 Abuse / misuse scenarios

ID	Scenario	Expected behaviour / acceptance
A1	A user tries to check in using another patient's booking reference.	Identity/authorization check fails; no queue ticket is issued; attempt is logged without exposing sensitive patient information.
A2	The same booking is presented at two kiosks.	Only one valid check-in is accepted; subsequent attempts are rejected or flagged.
A3	A user repeatedly creates bookings to block capacity.	Rate/booking policy limits or staff review should prevent uncontrolled reservation hoarding.
A4	An unauthorised staff account attempts to view patient bookings outside its role.	Access is denied and the event is auditable.
A5	A user manipulates a ticket number or queue state.	Queue state is server-controlled; client-side changes cannot directly alter another patient's status.

9. Counterexamples, Risks and Limitations

The strongest counterexample to the proposal is that a flexible arrival window can move the congestion problem rather than solve it. If too many patients are assigned the same window and most arrive near its start, the resulting queue may be longer than a well-designed fixed-slot system. The modelling literature explicitly shows that appointment scheduling-window choice creates trade-offs involving waiting and provider idle time [8].

A second counterexample is that online booking may not be the preferred or accessible route for every patient. The West China Hospital survey reported barriers related to capability and health-card requirements, while the Shanghai survey found low use of the studied web-based channel and continued support for on-the-spot options [5,7]. Therefore the proposed system must be additive: it should improve coordination for patients who can use it without making other routes unavailable.

A third limitation is generalisability. The strongest numerical evidence comes from large tertiary or general hospitals and may not transfer directly to small private clinics, community clinics,

specialist practices or different cities. For this reason, the project should not state that a 15% no-show rate, or any other literature value, is the baseline for all Chinese clinics.

A fourth limitation is that no-show identification does not automatically recover capacity. A booking that is never checked in can be recognised, but the clinic still needs a fair operational rule for releasing the slot or using a waitlist. This policy is outside pure interface design and must be agreed with the clinic.

Finally, the queue cannot be treated as a purely technical FIFO list. Clinical urgency, appointment type and doctor workflow may determine the order in which patients are called. The software should therefore record status and support authorised staff decisions rather than automate clinical prioritisation.

10. Proposed Evidence and Evaluation Method for the Semester Project

The project will use a mixed evidence strategy: public literature and policy evidence for context, a reproducible synthetic workflow model for early evaluation, and direct observation/interview evidence only if appropriate access and permission become available. No private patient data are required for the semester prototype.

Phase	Method	Output	Decision supported
1. Workflow validation	Observe or document the target clinic process; compare with the published workflow model.	Current-state process map and validated stakeholder responsibilities.	Whether the problem is present in the target context.
2. Baseline measurement	Collect or simulate appointment, arrival, no-show and queue timestamps using a declared data-generation method.	Baseline metrics and distributions.	Which pain points are measurable.
3. Alternative analysis	Compare proposed software with manual time-window + manual check-in/waitlist policy.	Trade-off matrix.	Whether software adds enough value to justify the semester intervention.
4. Prototype	Implement booking, window allocation, check-in, queue state and exception policies.	Executable prototype and tests.	Whether the workflow is implementable.
5. Controlled evaluation	Run the same synthetic scenarios through baseline and proposed workflows; freeze seeds and versions.	Reproducible results, failed runs and negative cases.	Whether the proposed workflow improves selected measures.
6. Reproduction	Publish code, environment, evidence ledger and instructions in a versioned repository.	Tagged evidence package.	Whether an independent evaluator can reproduce the analysis.

11. AI-Use and Source-Verification Log

AI was used as a research and writing assistant, not as a source of empirical facts. The final evidence claims in this brief were checked against the cited publisher/NCBI records and the National Health Commission pages. No AI-generated statistic was accepted without a traceable source.

Date	AI tool / task	Use	Accepted advice	Rejected / constrained advice	Verification
2 Oct 2026	ChatGPT - problem framing	Explored the clinic appointment problem and converted the initial idea into a testable workflow.	Focus on the booking-to-arrival state transition and on explicit normal/boundary/abuse cases.	Do not present assumptions such as “manual appointment systems are common everywhere” as facts.	Checked proposed claims against peer-reviewed Chinese studies and NHC guidance.
2 Oct 2026	ChatGPT - source discovery	Located public evidence on waiting, appointment systems, no-shows and national outpatient guidance.	Use at least two independent source types and include counterevidence.	No interviews, patient records or local statistics were invented to strengthen the argument.	Source metadata and findings checked on PubMed/PMC/Springer and NHC pages.
2 Oct 2026	ChatGPT - document drafting	Structured the evidence brief and repository/reproducibility materials.	Use a bounded scope, explicit assumptions, measures and stop conditions.	No unsupported efficacy claim, universal ranking, or fabricated experiment was included.	Every numerical claim in the report maps to an identified reference or is labelled illustrative.

12. Reproducibility and Versioned Artefact Plan

The associated repository package contains the evidence ledger, source notes, an executable evidence-summary notebook, the report and reproducibility instructions. The analysis notebook uses deterministic operations over the versioned CSV evidence ledger; no network access is required to rerun the summary.

Artefact	Purpose	Reproducibility note
report/ISE-A1_Problem_and_Stakeholder_Evidence_Brief.docx	Human-readable report.	Versioned with the same repository tag.
report/ISE-A1_Problem_and_Stakeholder_Evidence_Brief.pdf	PDF rendering for submission/review.	Generated from the DOCX and checked visually after rendering.
evidence/evidence_ledger.csv	Structured evidence records and extracted claims.	Stable rows/IDs E1-E8; no hidden external dependency.
evidence/source_notes.md	Source URLs, citation details and verification notes.	Contains the direct public source links used in the report.
notebook/evidence_summary.ipynb	Deterministic summary of evidence categories and quantitative reference points.	Runs top-to-bottom using the checked-in CSV.
src/README.md	Documents source organisation and future implementation boundaries.	No credentials or patient data included.

Artefact	Purpose	Reproducibility note
requirements.txt	Environment declaration.	Python package versions used by the notebook.
README.md	Reproduction and submission instructions.	Identifies the evidence package version/tag.

Version identifier: This prepared evidence package is tagged as “ise-a1-v1.0” in the local repository. A GitHub remote URL is intentionally not fabricated; the repository must be pushed to the student's own GitHub account before platform submission.

13. Conclusion

The evidence supports the existence of a meaningful outpatient appointment-coordination problem in China, but it also shows that the problem is nuanced. Appointment services can reduce registration friction and waiting, yet online booking does not remove no-shows or accessibility barriers. National policy already encourages timed appointment, identity-aware booking, no-show management and online/offline integration [3,4,6].

The proposed two-stage workflow responds to a specific weakness in the booking-to-arrival transition. Instead of treating a reservation as proof that a patient is present, the system makes arrival a separate, observable state through check-in. A flexible two- or three-window schedule is then used to make arrival less rigid, while the clinic dashboard uses confirmed arrivals to manage the queue. The idea is technically feasible at semester scale, but its benefit depends on capacity planning, patient adoption, accessibility, and operational policies for late arrivals and no-shows.

The appropriate A1 conclusion is therefore a conditional proceed-to-prototype decision. The next engineering step is not to build a full clinic platform. It is to validate the selected workflow, quantify its baseline, test alternative window sizes against arrival variability, and determine whether the additional software capability produces measurable value over a simpler operational change.

References

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Appendix A. ISE-A1 Requirement Coverage Map

Assignment requirement	Where addressed in this package
Investigate an authentic workflow using at least two independent source types.	Sections 2-3: official NHC policy/statistics + peer-reviewed Chinese empirical studies.
Identify users, affected parties, decision makers, constraints, and current responsibility.	Section 4: stakeholder table + current-state and proposed workflow.
Quantify pain, baseline, value measure, and failure cost.	Section 5: published reference baselines, local measurement definitions, and failure-cost analysis.
Compare software intervention with at least one non-software alternative.	Section 6: operational-change alternative using existing channels and manual check-in/waitlist.
Bound semester scope, non-goals, assumptions, and stop conditions.	Section 7.
Evidence ledger.	Section 3 + evidence/evidence_ledger.csv.
Problem/value statement.	Sections 1-2 and the A1 engineering-decision callout.
Stakeholder/workflow map.	Section 4.
Normal, boundary, abuse scenarios.	Section 8.
AI-use and source-verification log.	Section 11.
Reproducibility, failed/negative evidence and provenance.	Sections 9-12 + repository README and source notes.